

Physiology of Muscle contraction

Dr. Wafa Mubarak



Types of muscle tissue

- Striated – skeletal muscles
- – cardiac muscle
- Smooth



Physiological properties of muscles

- Excitability ,Conductivity , Contractility
- Automaticity (smooth & cardiac muscles)

Physical properties of muscles

- Elasticity (for striated)
- Plasticity (for smooth)



Functions of skeletal muscle

- Body movement
- Maintenance of posture
- Support soft tissue
- Role in regulation of body temperature



Muscles are biological machines

Muscles are biological machines made up of proteins.

The function of muscle is to convert chemical energy into mechanical work and force.

The chemical energy is obtained from ATP and CP



Chemical composition

Water

- Muscle contains about 75-80% of water.
- Water provides a good medium for inorganic and organic compounds.
- Water reduces friction and dehydration of muscles.

Proteins

- Muscle consists 3 types of proteins:- structural proteins, contractile proteins and enzymatic proteins

Minerals

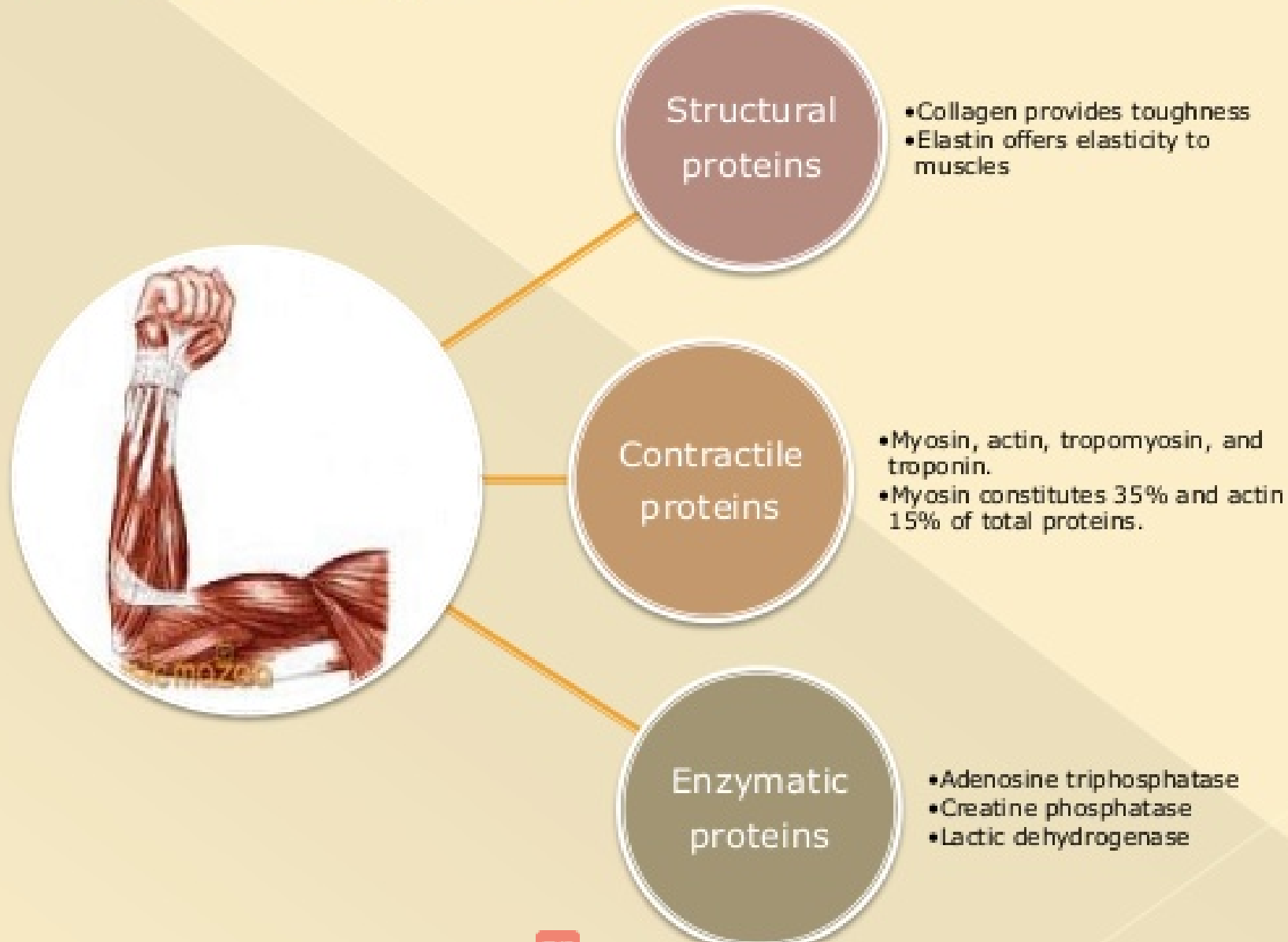
- Calcium ions of sarcoplasm initiates muscle contraction.
- Magesium ions are important for muscle contraction.
- sodium and Potassium ions set the action potential.

Organic compounds

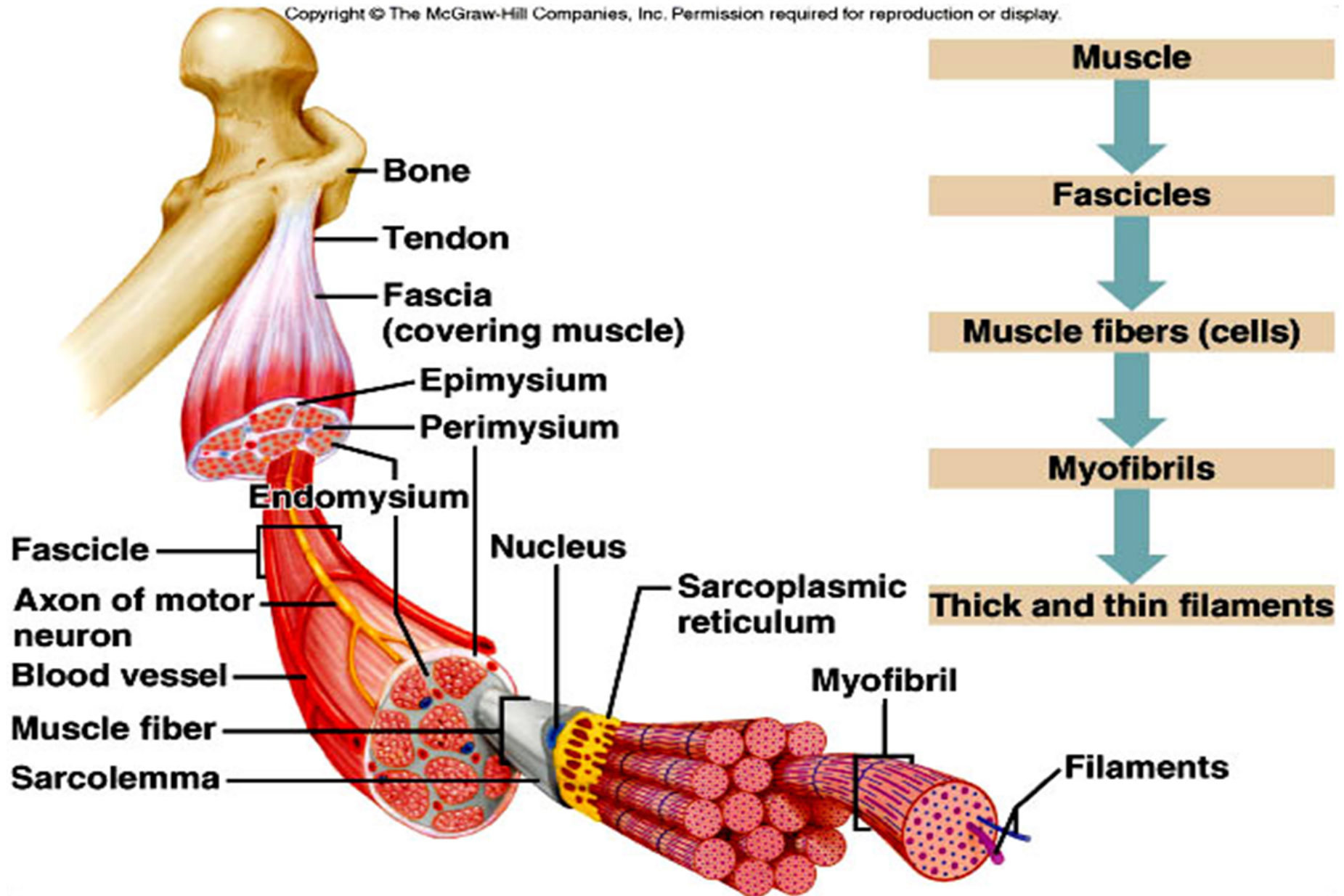
- Muscle is a storehouse of glycogen and oxidation of glycogen provides energy.
- The activity of muscle is proportional to the amount of phospholipids.
- ATP is the primary source of energy for muscle contraction.



Muscle proteins

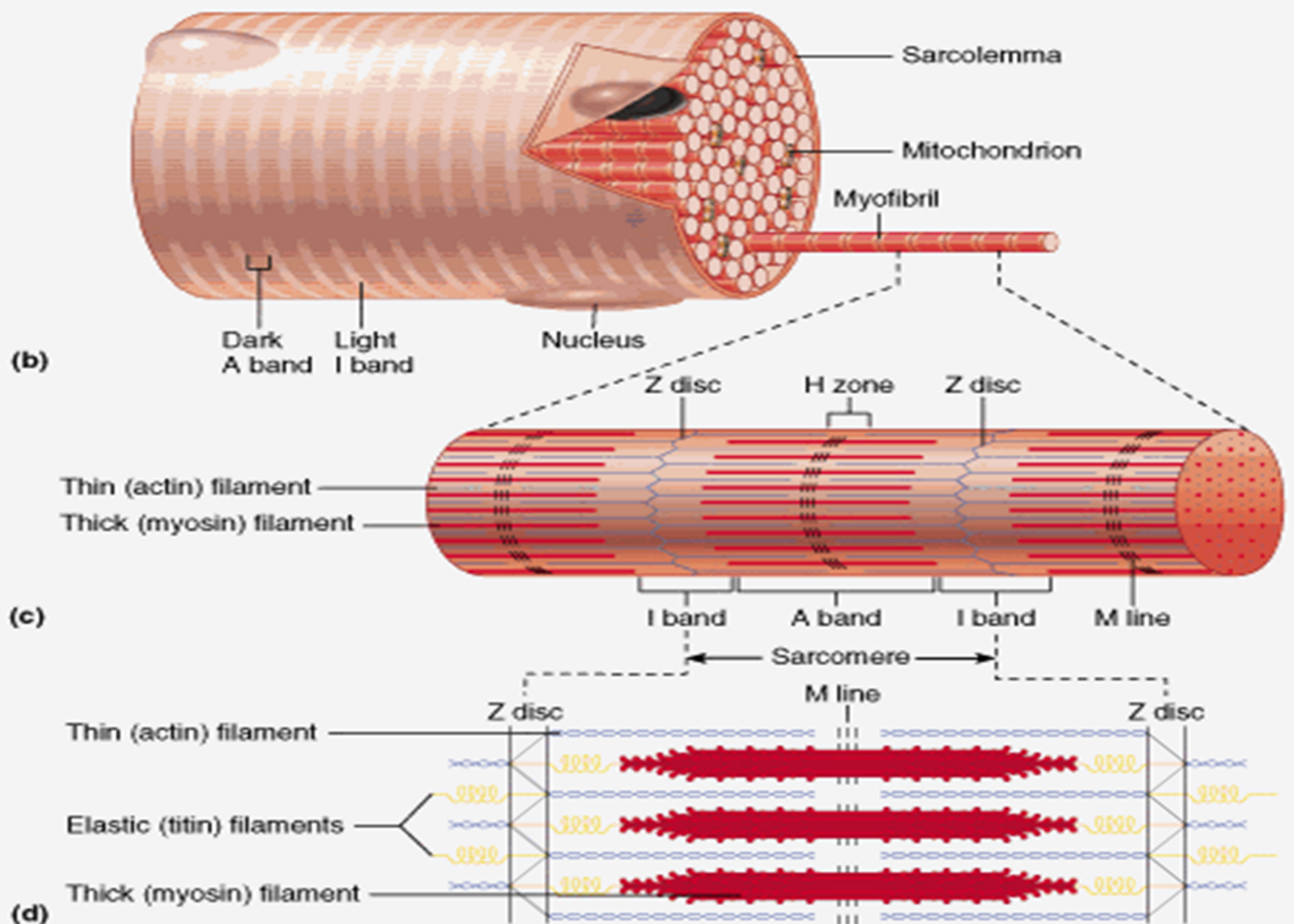


Structure of skeletal muscle



Skeletal Muscle Fibers

- Each muscle fiber is a single, long, cylindrical muscle cell.
- It's covered by **Sarcolemma**-cell membrane
- It has myofibrils which separated into compartments called **sarcomeres**(is the functional unit of the muscle) that contain :
 - 1- thick filaments (made up of the **myosin**)
 - 2- thin filaments (made up of the **actin**)



Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

A sarcomere extends from Z line to Z line, it contain:

a. I bands

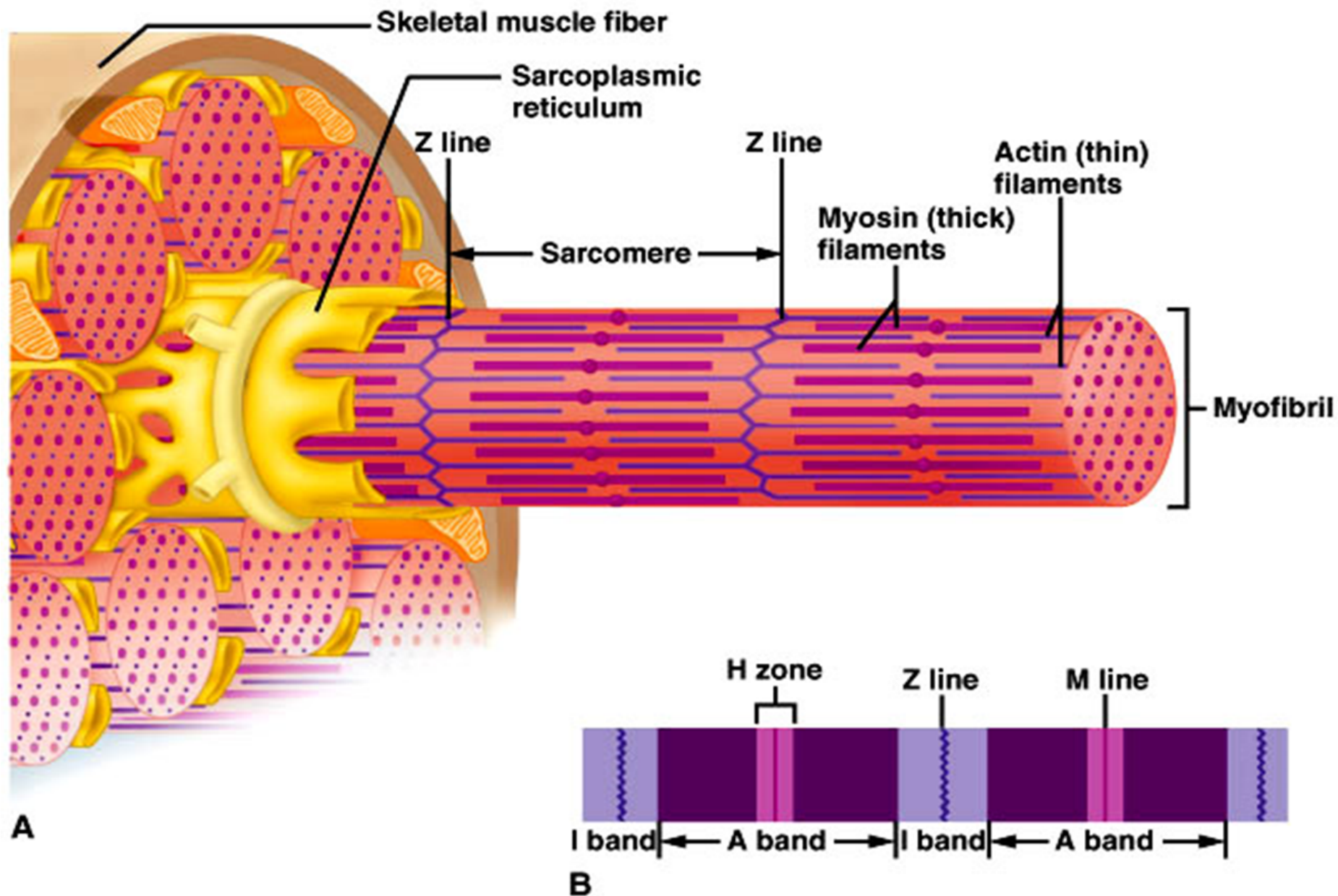
- light bands
- made up of actin

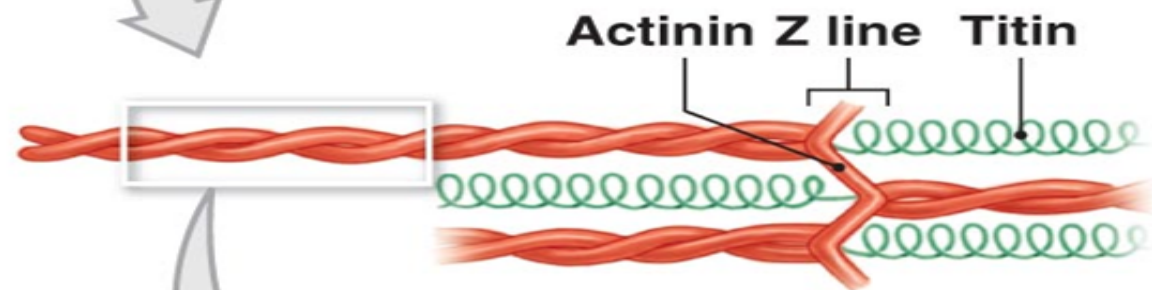
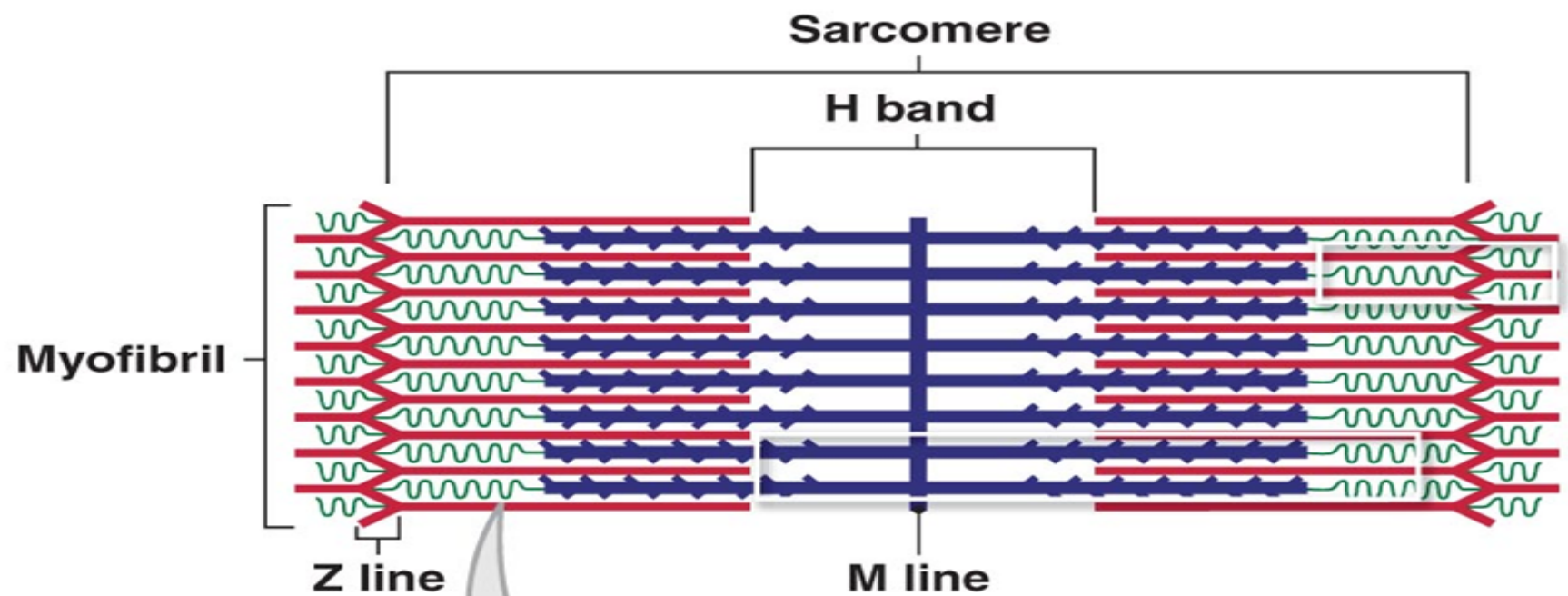
b. A bands

- dark bands
- made up of overlapping thick and thin filaments.(actin + myosin)

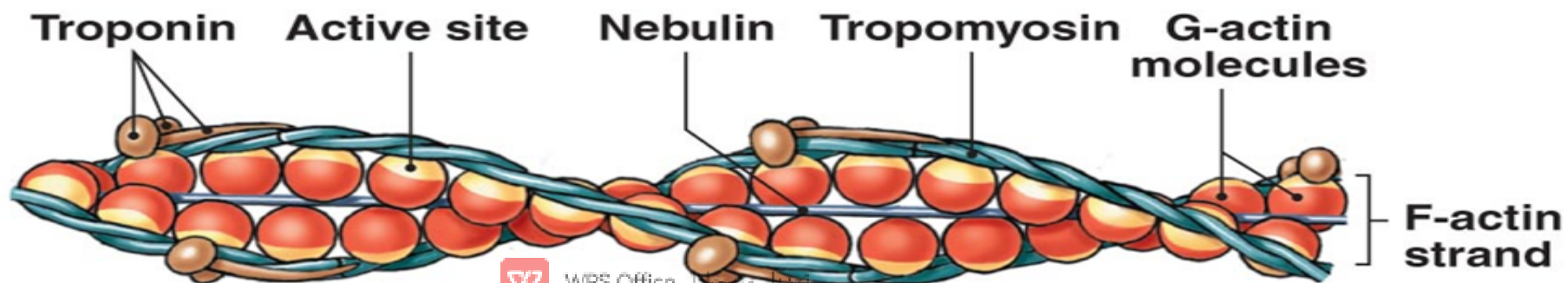
c. H zone In the center of A bands, consisting of myosin filaments only.



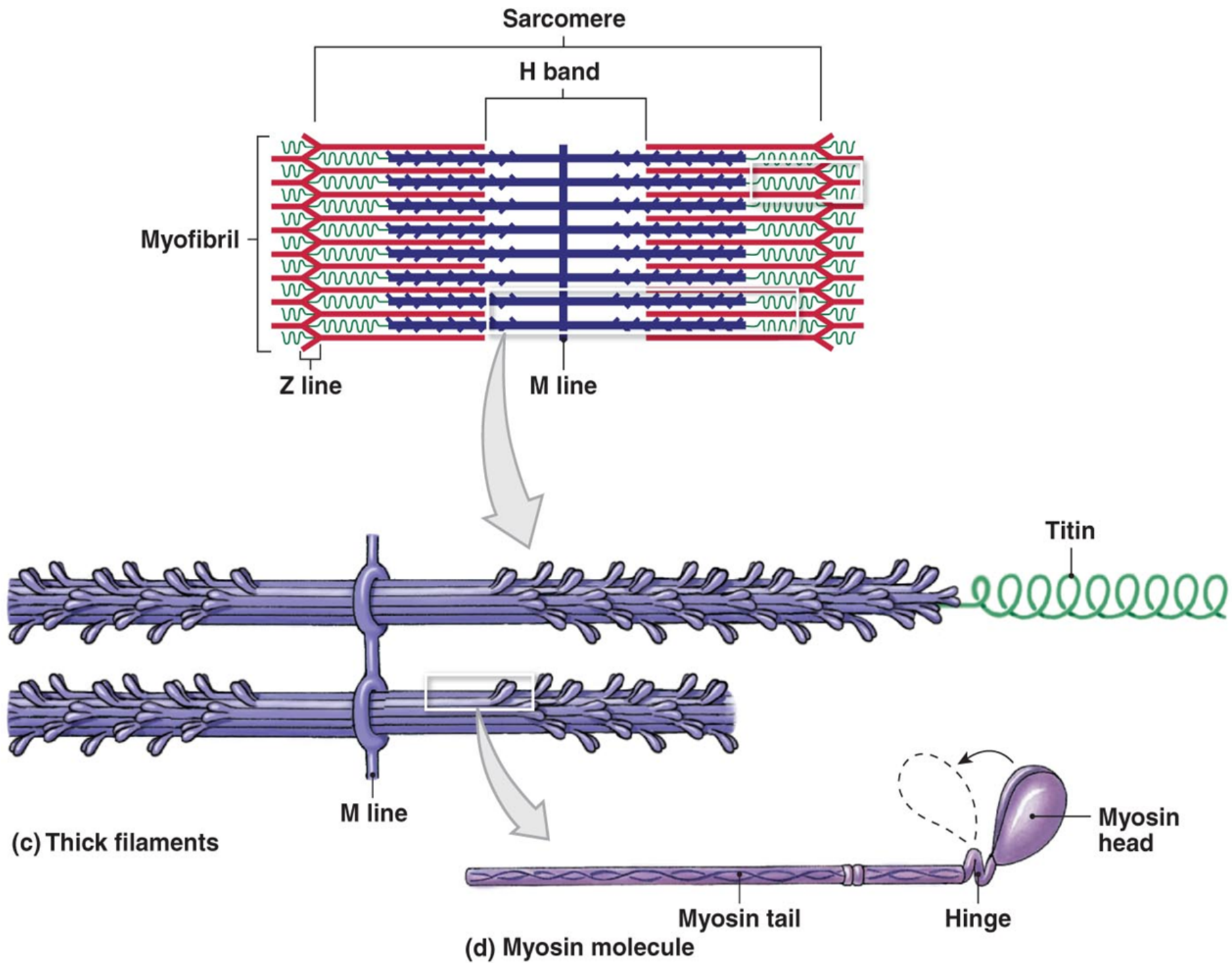




(a) Z line and thin filaments



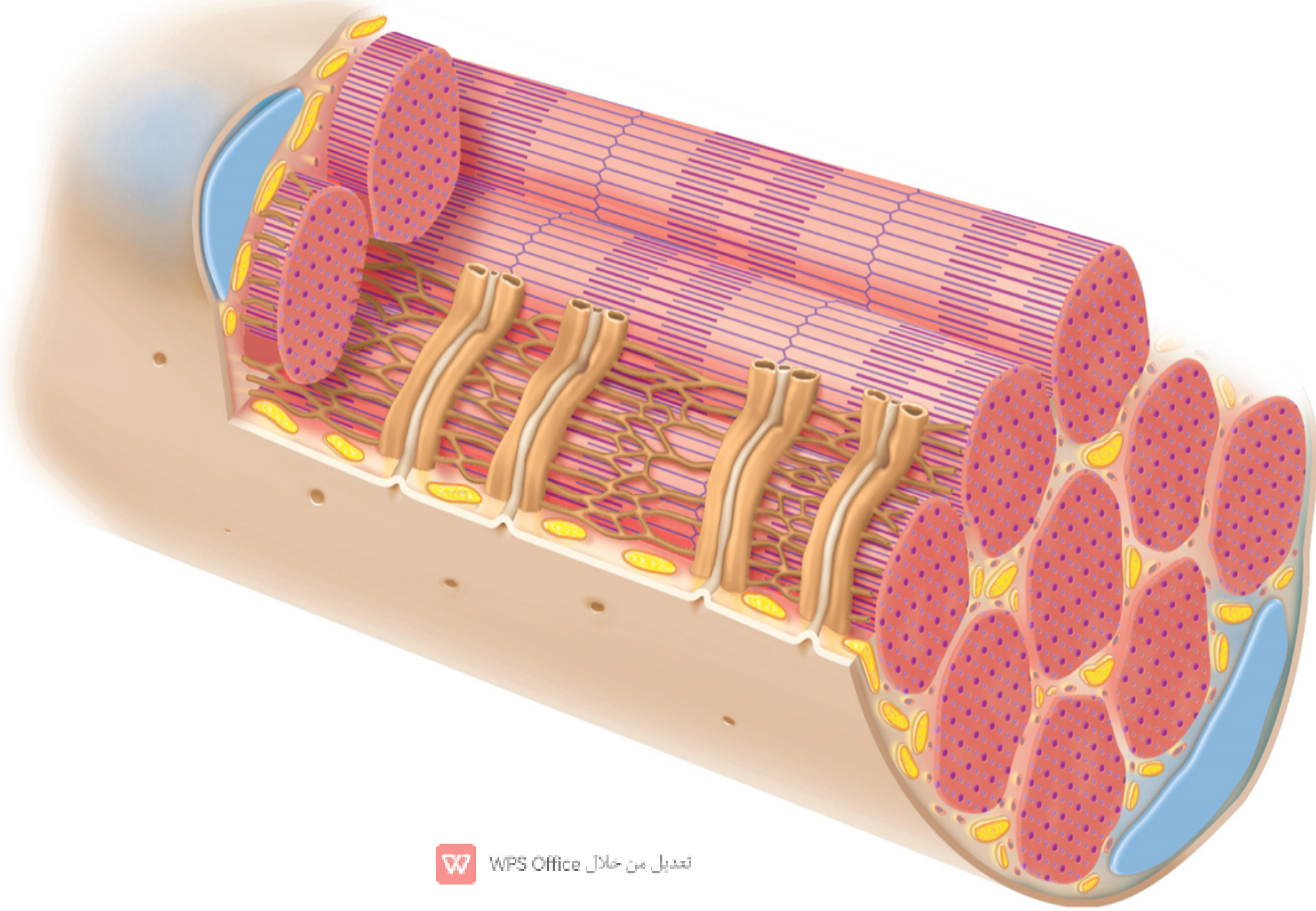
(b) Thin filament



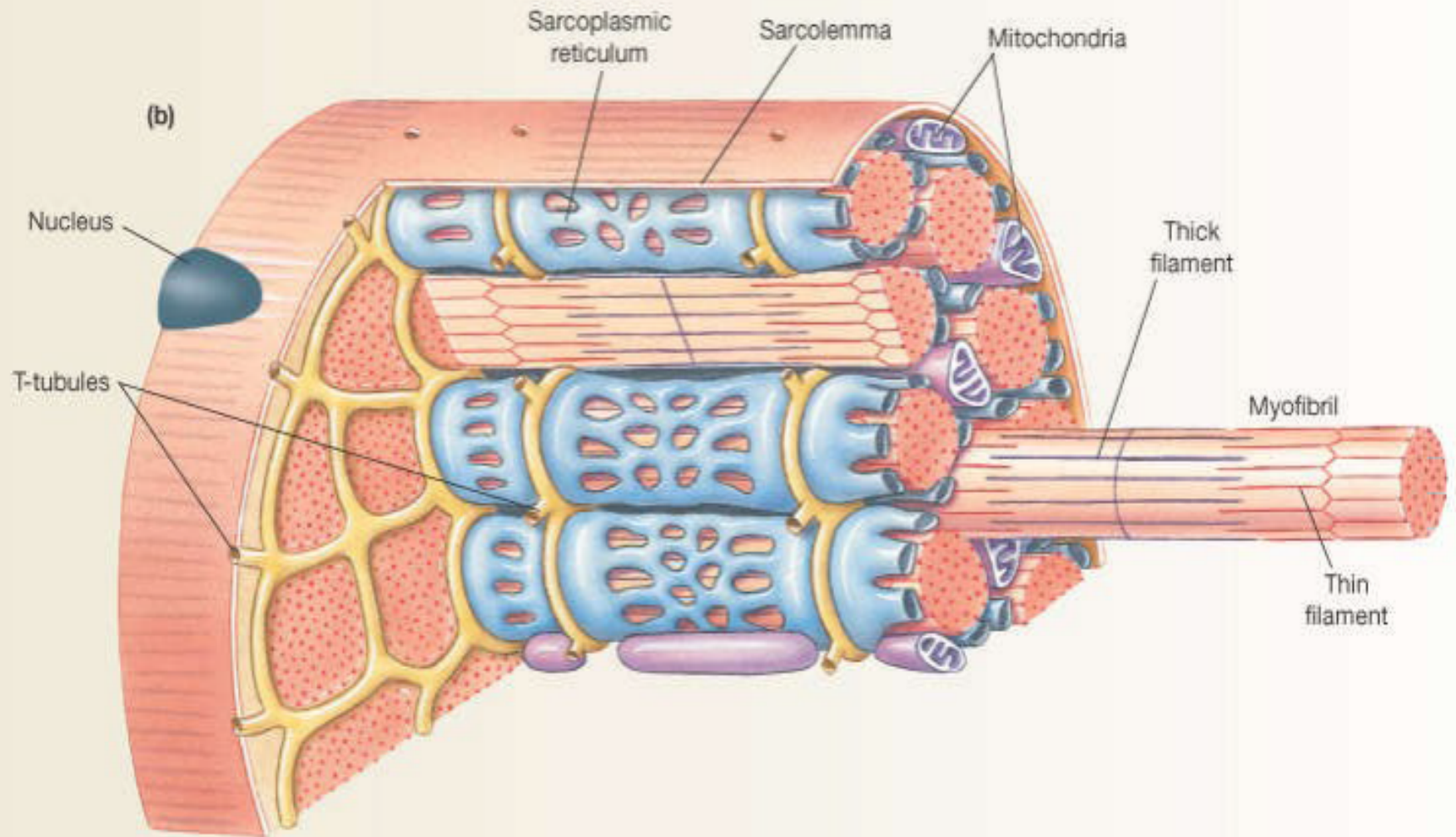
- Beneath the sarcolemma of a muscle fiber lies the sarcoplasmic reticulum (endoplasmic reticulum)
- which is associated with transverse (T) tubules (membrane channels)
- activate the muscle contraction mechanism



Sarcoplasmic Reticulum and Transverse Tubules



ULTRASTRUCTURE OF MUSCLE FIBER



Skeletal Muscle Contraction

2 events take place in a muscle contraction

1-shortening of sarcomeres

2-pulling of the muscle against its attachments.



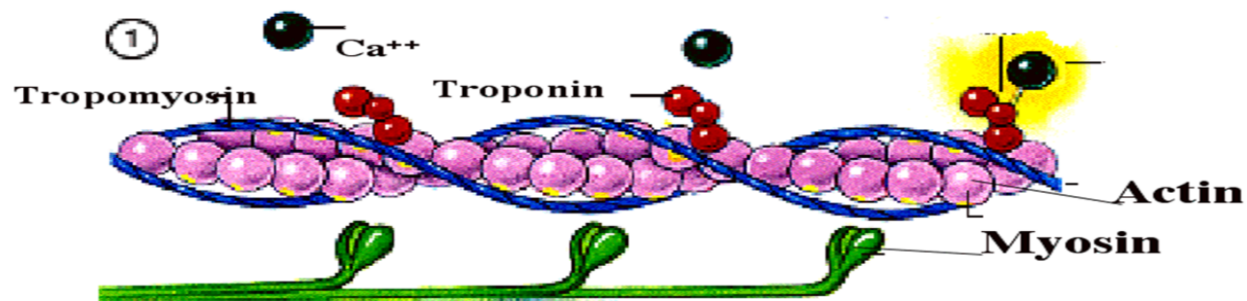
Role of Myosin and Actin

1. Myosin consists of two twisted strands

2. Actin is a globular protein with myosin binding sites;

-tropomyosin and troponin are two proteins associated with the surface of the actin filaments.

Actin with tropomyosin and troponin form the thin filament



Muscle Contraction—Steps

- 1-A nerve impulse enters the presynaptic terminal (nerve) of the neuromuscular junction.
- 2-The impulse causes Ach to be released from the synaptic vesicles in the axon terminal.
- 3- Ach diffuses across the synaptic cleft and opens Na^+ channels in muscle membranes.
- 4- Na^+ enters the muscle cell and depolarizes it.
- 5-“T” tubules carry impulses into the sarcoplasmic reticulum and releases Ca^{2+} ions.



6- Ca^{+2} enters the muscle fibrils and binds to troponin molecules on tropomyosin strands moving the strand and exposing the binding sites.

7- Myosin binds to actin forming cross-bridges that ATP can bind to.

8- ATP breaks down, releasing energy, causing cross bridges to pull actin strand.

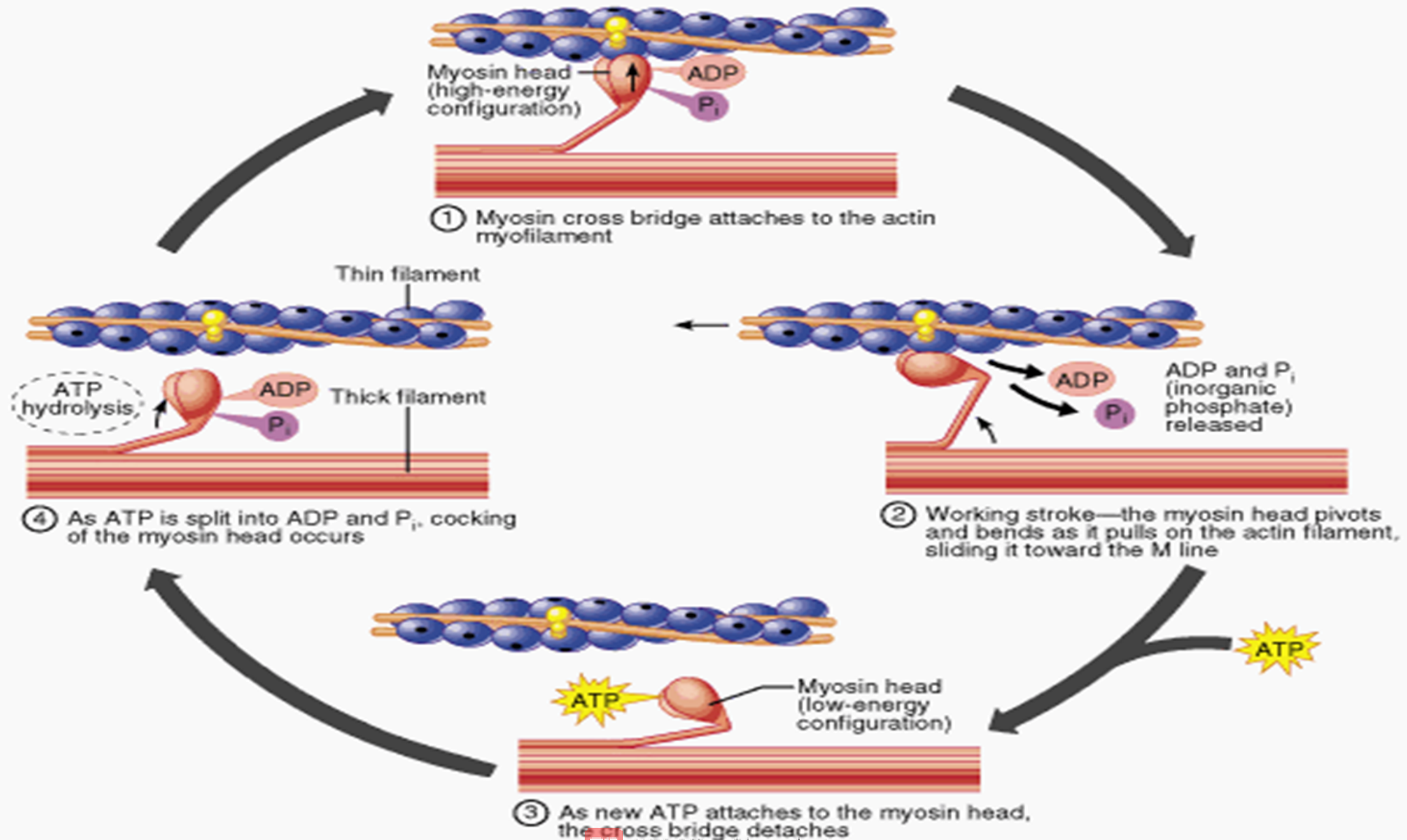


9- Another ATP binds to myosin cross bridge for the recovery stroke. (bend, attach, and pull) on the actin strand.

10- When the action potential ends Ca^{+2} ions are pumped back into the sarcoplasmic reticulum. Tropomyosin covers the binding sites and myosin can no longer bind.



sliding filament theory of muscle contraction



Muscle Relaxation

- This occurs when Ca^{++} is removed from mycoplasm back into sarcoplasmic reticulum by Ca^{++} pump located in the sarcoplasmic reticulum
- When Ca^{++} concentration is decreased:
 - Troponin returns to original state
 - Tropomyosin covers myosin binding sites
 - Cross-bridge cycling stops



Rigor mortis (postmortem rigidity)



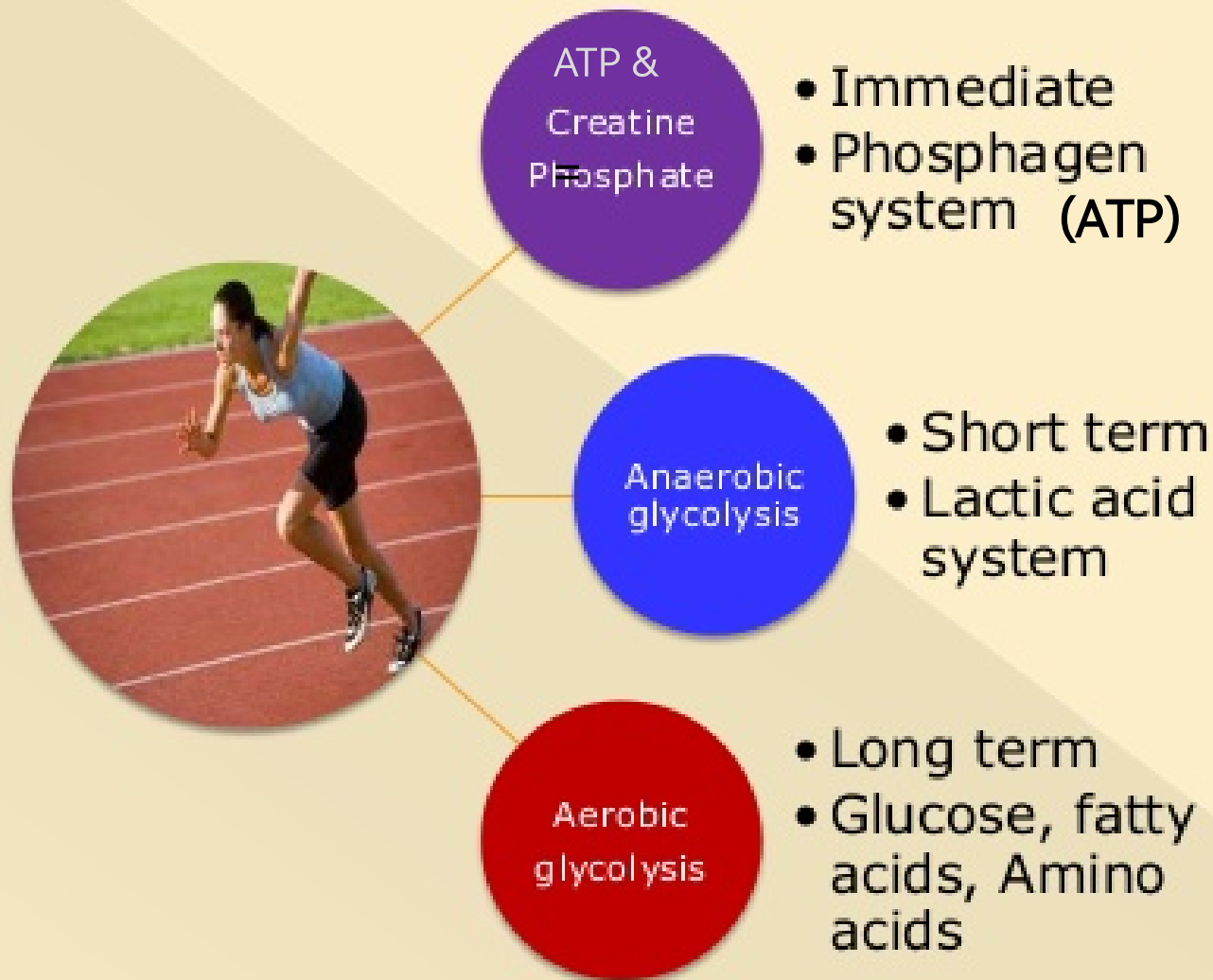
Energy expenditure during muscle contraction

- Na/K pump work
- Sliding of myosin heads along actin filaments
- Calcium pump work

**MOST ENERGY IS SPENT FOR
RELAXATION**



Energy for muscle contraction



Adenosine triphosphate, ATP

- ⦿ ATP is the immediate source of energy for muscle contraction.
- ⦿ The break down of phosphate bond of ATP releases maximum energy.

Anaerobic glycolysis:

Glucose $\rightarrow$ 2 moles of lactic acid + 8ATPs.

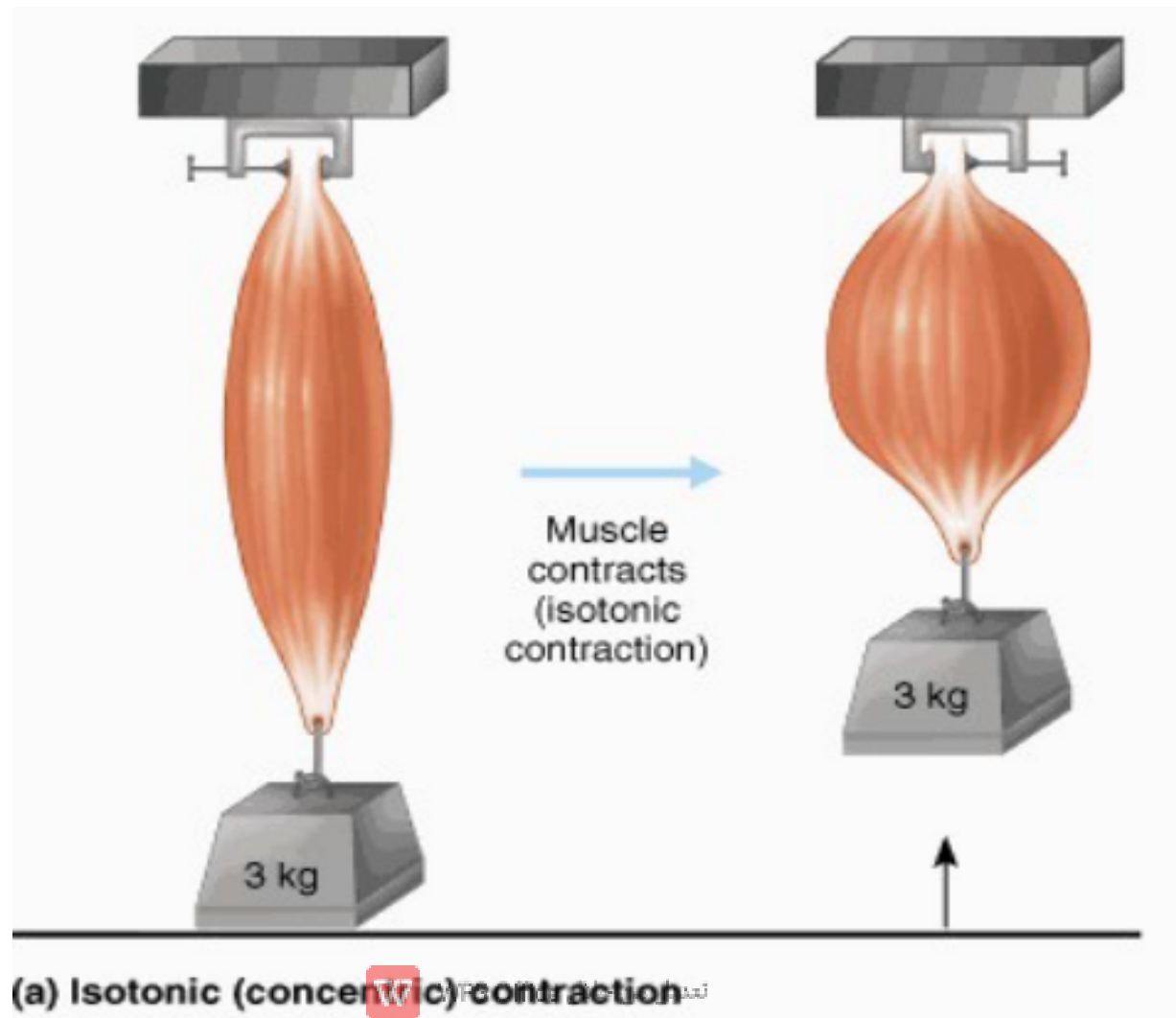
Aerobic glycolysis coupled with Krebs's cycle:

Glucose $\rightarrow$ 6 CO₂ + 6H₂O + 38 ATPs.



Types of skeletal muscle contraction

1-Isotonic contraction



- Isometric contraction

